

CUET PG (SCQP09) PRACTICE QUESTION BANK

Computer Science & Information Technology | Subject: Data Structures

TOPIC: STACKS (EASY LEVEL)

50 MCQs

MAX MARKS: 200

Instructions:

- Each question carries 4 marks. For each incorrect response, 1 mark will be deducted (-1).
- Select the single best answer choice for each multiple-choice question.
- Answer Key is provided on the final page of this question sheet.

Q1. A stack is a linear data structure that follows which principle?

- (A) FIFO (First In First Out)
- (B) LIFO (Last In First Out)
- (C) LILO (Last In Last Out)
- (D) Random Access

Q2. Which operation is used to insert an element into a stack?

- (A) Pop
- (B) Enqueue
- (C) Push
- (D) Peek

Q3. Which operation removes and returns the top element from a stack?

- (A) Push
- (B) Pop
- (C) Peek
- (D) Dequeue

Q4. Which operation returns the top element of the stack without removing it?

- (A) Pop
- (B) Push
- (C) Peek / Top
- (D) Seek

Q5. Attempting to pop an element from an empty stack results in:

- (A) Stack Overflow
- (B) Stack Underflow
- (C) Garbage Value
- (D) Segmentation Fault

Q6. Attempting to push an element into a completely full stack results in:

- (A) Stack Overflow
- (B) Stack Underflow
- (C) Memory Leak
- (D) Null Pointer Exception

Q7. What is the initial value of the top pointer when a stack is implemented using an array (0-indexed) and is empty?

- (A) 0
- (B) -1
- (C) 1
- (D) NULL

Q8. What is the time complexity of the Push operation in a standard array-based stack?

- (A) $O(1)$
- (B) $O(N)$
- (C) $O(\log N)$
- (D) $O(N^2)$

Q9. What is the time complexity of the Pop operation in a stack?

- (A) $O(N)$
- (B) $O(1)$
- (C) $O(\log N)$
- (D) $O(N \log N)$

Q10. Which real-world structure best models the working of a stack?

- (A) Queue at a ticket counter
- (B) Stack of plates at a buffet
- (C) People standing in a line
- (D) Traffic on a one-way road

- Q11. When implementing a stack using a singly linked list, push and pop operations should ideally be performed at:**
- (A) The end of the list
 - (B) The beginning (head) of the list
 - (C) The middle of the list
 - (D) Any random position
- Q12. What condition checks if an array-based stack of size `MAX` (0-indexed) is full?**
- (A) $\text{top} == \text{MAX}$
 - (B) $\text{top} == \text{MAX} - 1$
 - (C) $\text{top} == 0$
 - (D) $\text{top} == -1$
- Q13. In a linked list implementation of a stack, what does an empty stack condition correspond to?**
- (A) $\text{top} == \text{NULL}$
 - (B) $\text{top} \rightarrow \text{next} == \text{NULL}$
 - (C) $\text{top} == 0$
 - (D) $\text{top} \rightarrow \text{data} == -1$
- Q14. Which data structure is implicitly used by compilers to manage function calls and recursion?**
- (A) Queue
 - (B) Call Stack
 - (C) Heap
 - (D) Binary Tree
- Q15. In expression evaluation, what is another name for Polish Notation?**
- (A) Infix Notation
 - (B) Prefix Notation
 - (C) Postfix Notation
 - (D) Mixed Notation
- Q16. What is Reverse Polish Notation (RPN)?**
- (A) Infix Expression
 - (B) Prefix Expression
 - (C) Postfix Expression
 - (D) Binary Notation
- Q17. In which notation is the operator placed between the operands (e.g., $A + B$)?**
- (A) Postfix
 - (B) Prefix
 - (C) Infix
 - (D) Polish
- Q18. What is the postfix form of the infix expression $A + B$?**
- (A) $+ AB$
 - (B) $AB +$
 - (C) $A B +$
 - (D) $+ A B$
- Q19. What is the prefix form of the infix expression $A * B$?**
- (A) AB^*
 - (B) $*AB$
 - (C) $A*B$
 - (D) $*BA$
- Q20. Which feature in text editors is commonly implemented using a stack?**
- (A) Find & Replace
 - (B) Undo / Redo operations
 - (C) Spell Check
 - (D) Text Formatting
- Q21. To check for balanced parentheses in an expression, which data structure is most appropriate?**
- (A) Queue
 - (B) Stack
 - (C) Array
 - (D) Graph
- Q22. What happens to the top pointer when an element is pushed into an array-based stack?**
- (A) top is decremented
 - (B) top is incremented
 - (C) top remains unchanged
 - (D) top becomes 0

- Q23. What happens to the top pointer when an element is popped from an array-based stack?**
- (A) top is incremented (B) top is decremented
(C) top becomes -1 (D) top is doubled
- Q24. If elements A, B, C are pushed into a stack in order (A first, then B, then C), in what order will they be popped?**
- (A) A, B, C (B) C, B, A
(C) B, A, C (D) C, A, B
- Q25. Evaluate the following postfix expression: `5 3 2 \* +`**
- (A) 16 (B) 11
(C) 13 (D) 25
- Q26. What is the result of evaluating the postfix expression `8 2 / 3 +`?**
- (A) 7 (B) 4
(C) 10 (D) 5
- Q27. When converting an infix expression to postfix using a stack, what is stored in the stack?**
- (A) Operands (B) Operators and Parentheses
(C) Final Result (D) Variable names
- Q28. When evaluating a postfix expression using a stack, what is stored in the stack?**
- (A) Operators (B) Operands / Intermediate Numerical Values
(C) Parentheses (D) Code statements
- Q29. What is the postfix equivalent of `A + B \* C`?**
- (A) AB+C* (B) ABC*+
(C) A*BC+ (D) ABC+*
- Q30. What is the postfix equivalent of `(A + B) \* C`?**
- (A) AB+C* (B) ABC*+
(C) AB*C+ (D) +AB*C
- Q31. Which of the following operator has the highest precedence during operator-stack processing?**
- (A) + (B) -
(C) * (D) ^ (Exponentiation)
- Q32. In the back-button navigation mechanism of web browsers, which data structure is used?**
- (A) Queue (B) Stack
(C) Tree (D) Priority Queue
- Q33. Minimum number of stacks required to implement a FIFO Queue is:**
- (A) 1 (B) 2
(C) 3 (D) 4
- Q34. How many queues are minimum needed to implement a stack?**
- (A) 1 (B) 2
(C) 3 (D) 4

- Q35. In a stack implemented using a dynamic array, when full capacity is reached, what is typical resizing factor?**
- (A) Add 1 (B) Double the size (2x)
(C) Triple the size (3x) (D) Square the size
- Q36. Which algorithm graph traversal technique uses a Stack data structure explicitly or recursively?**
- (A) Breadth First Search (BFS) (B) Depth First Search (DFS)
(C) Prim's Algorithm (D) Kruskal's Algorithm
- Q37. What is the prefix expression for $(A - B) / C$?**
- (A) $-ABC$ (B) $-AB C$
(C) $- A B C$ (D) $AB-C/$
- Q38. What is the value of `top` when a stack of capacity 5 contains 3 elements (0-indexed implementation)?**
- (A) 3 (B) 2
(C) 4 (D) 5
- Q39. Reversing a string using a stack requires how many passes over the string?**
- (A) 1 pass (B) 2 passes (one push pass, one pop pass)
(C) 3 passes (D) Log N passes
- Q40. If a stack is created with array size 100, what is the maximum number of items it can store?**
- (A) 99 (B) 100
(C) 101 (D) Unlimited
- Q41. Which operation of stack can cause Underflow?**
- (A) Push (B) Pop
(C) Peek (D) Display
- Q42. Which operation of stack can cause Overflow?**
- (A) Pop (B) Push
(C) Peek (D) IsEmpty
- Q43. Evaluate the prefix expression $+ * 2 3 4$:**
- (A) 14 (B) 10
(C) 20 (D) 24
- Q44. In a stack, elements are accessed from:**
- (A) Both ends (B) Only top end
(C) Only bottom end (D) Middle
- Q45. Which of the following statements is TRUE about stacks?**
- (A) Random access is allowed (B) Direct access to middle element is $O(1)$
(C) Access is restricted to top element (D) Elements are sorted automatically
- Q46. What is the auxiliary space complexity of reversing a stack using recursion?**
- (A) $O(1)$ (B) $O(N)$ due to call stack
(C) $O(N^2)$ (D) $O(\log N)$

Q47. Is dynamic implementation of stack using Linked List subject to Stack Overflow (in theoretical memory context)?

- (A) Yes, always (B) No, unless system memory is exhausted
(C) Yes, when top reaches 100 (D) Depends on compiler

Q48. Which operator associativity is right-to-left in standard expression parsing?

- (A) + (B) *
(C) ^ (Exponentiation) (D) /

Q49. When processing balanced parenthesis check for string ``{()}``, what is in stack when processing `)`?

- (A) `{, [, `(` (B) `}`
(C) Empty (D) `{` only

Q50. Standard C Library function `qsort()` or recursive function stack frame cleanup relies on:

- (A) Heap Memory Allocation (B) System Call Stack
(C) Priority Queue (D) Circular Queue

ANSWER KEY (Q1 - Q50)

Q.	Ans	Q.	Ans	Q.	Ans	Q.	Ans	Q.	Ans
1	B	11	B	21	B	31	D	41	B
2	C	12	B	22	B	32	B	42	B
3	B	13	A	23	B	33	B	43	B
4	C	14	B	24	B	34	B	44	B
5	B	15	B	25	B	35	B	45	C
6	A	16	C	26	A	36	B	46	B
7	B	17	C	27	B	37	C	47	B
8	A	18	B	28	B	38	B	48	C
9	B	19	B	29	B	39	B	49	A
10	B	20	B	30	A	40	B	50	B